ECE 5360

a) $V_T = \phi_b - \frac{\sigma_\pi}{C_b}$. $\quad C_b = \frac{\epsilon_b}{t_b} \Rightarrow V_T = \phi_b - \frac{\sigma_\pi t_b}{\epsilon_b} = \phi_b - \frac{\sigma_\pi t_b}{10\epsilon_0}$

Recall $\sigma_\pi/q_h = 1.5\cdot 10^{13}\ cm^{-2}$

$$\Rightarrow \boxed{\begin{array}{l} V_T(t_b = 7nm) = -0.9V \\ V_T(t_b = 20nm) = -4.428V \end{array}}$$

Getting zero, $\phi_b = \frac{\sigma_\pi t_b}{10\epsilon_0} \Rightarrow t_b = \frac{10\epsilon_0\phi_b}{\sigma_\pi} = \boxed{t_b = 3.685nm}$ at $V_T = 0$

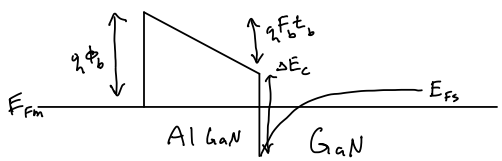
No, this is not ideal for an E-Mode (normally off) GaN HEMT. Since threshold voltage is zero, the device is on the edge of being normally off and normally on. The operating point is unstable, the device is unreliable and small variations/noise can cause it to turn on unintentionally.

b)

The threshold voltage in a GaN HEMT doesn't depend on cap layer thickness or conduction band offset, since they do not meaningfully affect polarization charges or electric field at the AlGaN/GaN interface. While thicker caps decrease gate capacitance, it doesn't affect the 2DEG density or electric field at the interface. Similarly, the band offset helps confine the 2DEG but doesn't meaningfully change the equilibrium at the interface either as long as it's sufficiently large. As such, VT is mostly independent of the cap properties and is mostly dependent on the barrier layer properties.

c) If you just set $t_c = 0$, the $\Delta E_c - \Delta E_c$ term remains, so no ΔE_c dependence. You can't just zero t_c since the heterojunction dynamics change completely with no cap.

Instead, for a single heterojunction AlGaN-GaN, it roughly looks like



$\Rightarrow$ Band geometry, we get:

$$E_{Fm} + q_h\phi_b - q_h F_b t_b - \Delta E_c + E_0 + (F_{Fs} - E_0) = E_{Fs}$$

Using the same substitutions

$$\frac{n_s}{n_b} + A n_s^{2/3} + kT\ln\left(e^{\frac{n_s}{N_{2D}}} - 1\right) = q_h V_{gs} - q_h\phi_b + \Delta E_c + q_h\frac{\sigma_\pi}{C_b}$$

(typo on handout)

dividing by kT and exponentiating, $\ldots = \exp\left(\frac{V_{gs} - \phi_b + \Delta E_c/q_h + \sigma_\pi/C_b}{V_{th}}\right)$

$$\Rightarrow \boxed{V_T = \phi_b - \frac{\sigma_\pi}{C_b} - \frac{\Delta E_c}{q_h}}$$

d) $C_g = \frac{\partial q_h n_s}{\partial V_{gs}}$

From eq 5, $\exp\left(\frac{q_h n_s}{C_c V_{th}} + \frac{q_h n_s}{C_b V_{th}} + \left(\frac{n_s}{n_0}\right)^{2/3}\right)\left(\underbrace{e^{\frac{n_s}{n_q}}}_{\approx e^{n_s/n_q}} - 1\right) = \underbrace{e^{\frac{V_{gs}-V_T}{V_{th}}}}_{\to V_{gs}}$ since $V_{gs} > V_T$

Since $V_{gs} > V_T$, $V_{gs} - V_T \approx V_{gs}$, $n_q = N_c^{2d} = \frac{C_q V_{th}}{q_h}$, $\underbrace{e^{\frac{n_s}{n_q}} - 1}_{\to e^{n_s/n_q}}$

↳ quantum capacitance.

$$\Rightarrow \quad \frac{q_h n_s}{C_c V_{th}} + \frac{q_h n_s}{C_b V_{th}} + \left(\frac{n_s}{n_0}\right)^{2/3} + \frac{q_h n_s}{C_q V_{th}} \approx \frac{V_{gs}}{V_T}$$

By chain rule, operate with $\frac{\partial}{\partial V_{gs}}$

$$\frac{\partial q_h n_s}{\partial V_{gs}}\left(\frac{1}{C_c V_{th}} + \frac{1}{C_b V_{th}} + \frac{2}{3}\frac{1}{q_h n_0}\left(\frac{n_s}{n_0}\right)^{-\frac{1}{3}} + \frac{1}{C_q V_{th}}\right) \approx \frac{1}{V_{th}}$$

Simplifying,

$$\underbrace{\frac{\partial q_h n_s}{\partial V_{gs}}}_{C_g}\left(\frac{1}{C_c} + \frac{1}{C_b} + \frac{2}{3}\frac{V_{th}}{q_h n_0^{2/3}} n_s^{-\frac{1}{3}} + \frac{1}{C_q}\right) \approx 1$$

$$C_g\left(\frac{1}{C_c} + \frac{1}{C_b} + \frac{1}{C_{q_h}} + \frac{2V_{th}}{3 q_h n_0^{2/3} n_s^{1/3}}\right) \approx 1$$

$$\Rightarrow \quad \boxed{C_g \approx \frac{1}{\frac{1}{C_c} + \frac{1}{C_b} + \frac{1}{C_{q_h}} + \frac{2}{3}\frac{V_{th}}{q_h n_s^{1/3} n_0^{2/3}}}}$$

Identical to expression given in problem as solution ✓

e)

$$C_g \approx \frac{1}{\frac{1}{C_c} + \frac{1}{C_b} + \frac{1}{C_{q_h}} + \cancel{\frac{2}{3}\frac{V_{th}}{q_h n_s^{1/3} n_0^{2/3}}}} \Rightarrow C_g \approx \left(\frac{1}{C_c} + \frac{1}{C_b} + \frac{1}{C_{q_h}}\right)^{-1}$$

neglect n_s dependence

Recall

$$\frac{I_d}{W} = q_h \mu V_{th} \frac{1}{L}\left[\left(\frac{1}{n_c} + \frac{1}{n_b} + \frac{1}{n_{q_h}}\right)\left(\frac{n_{s0}^2 - n_{sL}^2}{2}\right) + \frac{2}{5}\frac{n_{s0}^{5/3} - n_{sL}^{5/3}}{n_0^{2/3}} - n_{q_h}\left(e^{-\frac{n_{s0}}{n_{q_h}}} - e^{-\frac{n_{sL}}{n_{q_h}}}\right)\right]$$

Where $e^{\frac{n_{s0}}{n_c}} e^{\frac{n_{s0}}{n_b}} e^{\left(\frac{n_{s0}}{n_0}\right)^{2/3}}\left(e^{\frac{n_{s0}}{n_{q_h}}} - 1\right) \approx e^{\frac{V_{gs}-V_T}{V_{th}}}$ $\quad e^{\frac{n_{sL}}{n_c}} e^{\frac{n_{sL}}{n_b}} e^{\left(\frac{n_{sL}}{n_0}\right)^{2/3}}\left(e^{\frac{n_{sL}}{n_{q_h}}} - 1\right) \approx e^{\frac{V_{gs}-V_T-\cancel{V_{ds}}^{\,0}}{V_{th}}}$ to justify C_g approx

In the I_D/W expression we can neglect the e^{-x} terms as per the derivation, they relate to the -1 term in $e^{n_s/n_{q_h}} - 1$, which we already neglect

$$\frac{I_d}{W} = q_h \mu V_{th} \frac{1}{L}\left[\left(\frac{1}{n_c} + \frac{1}{n_b} + \frac{1}{n_{q_h}}\right)\left(\frac{n_{s0}^2 - n_{sL}^2}{2}\right) + \frac{2}{5}\frac{n_{s0}^{5/3} - n_{sL}^{5/3}}{n_0^{2/3}}\right]$$

From the n_{so}, n_{sL} equations $\Rightarrow \left(\frac{1}{n_c}+\frac{1}{n_b}+\frac{1}{n_q}\right)n_s + \frac{1}{n_0^{2/3}}n_s^{2/3} = \frac{V_{gs}-V_T}{V_{th}}$

Squaring both sides $\Rightarrow \left(\frac{1}{n_c}+\frac{1}{n_b}+\frac{1}{n_q}\right)^2 n_s^2 + \left(\frac{1}{n_c}+\frac{1}{n_b}+\frac{1}{n_q}\right)\frac{1}{n_0^{2/3}}n_s^{5/3} = \frac{(V_{gs}-V_T)^2}{V_{th}^2}$

(general form sub in n_{so}, n_{sL})

Recall $\frac{I_d}{W} = q_h \mu V_{th}\frac{1}{L}\left[\left(\frac{1}{n_c}+\frac{1}{n_b}+\frac{1}{n_q}\right)\left(\frac{n_{so}^2-n_{sL}^2}{2}\right) + \frac{2}{5}\frac{n_{so}^{5/3}-n_{sL}^{5/3}}{n_0^{2/3}}\right]$

$$\Rightarrow \frac{I_D}{W} = \frac{q_h\mu V_{th}}{2L}\left(\frac{1}{n_c}+\frac{1}{n_b}+\frac{1}{n_q}\right)^{-1}\underbrace{\left[\left(\frac{1}{n_c}+\frac{1}{n_b}+\frac{1}{n_q}\right)^2(n_{so}^2-n_{sL}^2) + \left(\frac{1}{n_c}+\frac{1}{n_b}+\frac{1}{n_q}\right)\frac{4}{5n_0^{2/3}}\left(n_{so}^{5/3}-n_{sL}^{5/3}\right)\right]}_{\approx \text{ roughly matches identity}}$$

Subbing and approximating our above identity roughly we finally arrive at

$$\frac{I_d}{W} \approx \frac{q_h\mu V_{th}}{2L}\left(\frac{1}{n_c}+\frac{1}{n_b}+\frac{1}{n_q}\right)^{-1}\left(\frac{V_{gs}-V_T}{V_{th}}\right)^2 = \mu\left[\frac{V_{th}}{q_h}\left(\frac{1}{n_c}+\frac{1}{n_b}+\frac{1}{n_q}\right)\right]^{-1}\frac{(V_{gs}-V_T)^2}{2L}$$

$$\approx \mu\underbrace{\left[\frac{1}{C_c}+\frac{1}{C_b}+\frac{1}{C_q}\right]^{-1}}_{C_g}\frac{(V_{gs}-V_T)^2}{2L} \approx \mu C_g\frac{(V_{gs}-V_T)^2}{2L}$$

$$\boxed{\frac{I_d}{W} \approx \mu C_g\frac{(V_{gs}-V_T)^2}{2L}}$$

f) For simplicity let W be $100\,\mu m$ (normal) and $t_c = 2nm$

Other params scale accordingly. $\boxed{W = 100\,\mu m}$ $\boxed{t_c = 2nm}$

We require $V_T = -4$.

Recall $V_T = \phi_b - \frac{\sigma_\pi t_b}{10\epsilon_0} \Rightarrow t_b = \frac{\phi_b - V_T}{\sigma_\pi}\cdot 10\epsilon_0 = 18.4nm$ $\boxed{t_b = 18.4nm}$

Note $C_q = \frac{q_h^2 m^*}{\pi\hbar^2} = \frac{8q_h}{hv_F} \approx 386\,\frac{aF}{\mu m}$ is quantum capacitance.

$$C_g \approx \left(\frac{1}{C_c}+\frac{1}{C_b}+\frac{1}{C_q}\right)^{-1} = \left(\frac{t_b}{10\epsilon_0}+\frac{t_c}{10\epsilon_0}+\left(\approx \overset{386\,\frac{aF}{\mu m}}{\frac{8q_h^2}{hv_F}}\right)^{-1}\right)^{-1}$$

We require $I_d = 1$ amp at $V_{gs} = 0.5V$, $V_T = -4V$

$\Rightarrow I_d/W \approx \mu C_g\frac{(V_{gs}-V_T)^2}{2L} \Rightarrow L \approx W\mu C_g\frac{(V_{gs}-V_T)^2}{2I_d} \approx 5.4\mu m$

↳ defined above. μm

All params specified (same other params as in q_h)

$\boxed{W = 100\,\mu m}$ $\boxed{t_c = 2nm}$ $\boxed{t_b = 18.4nm}$ $\boxed{L - 5.4}$